WORKED SOLUTION

STEP III 1994, Question 11

Mechanics • Equilibrium of Rigid Bodies

STEP III 1994, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s3-q11

The key idea

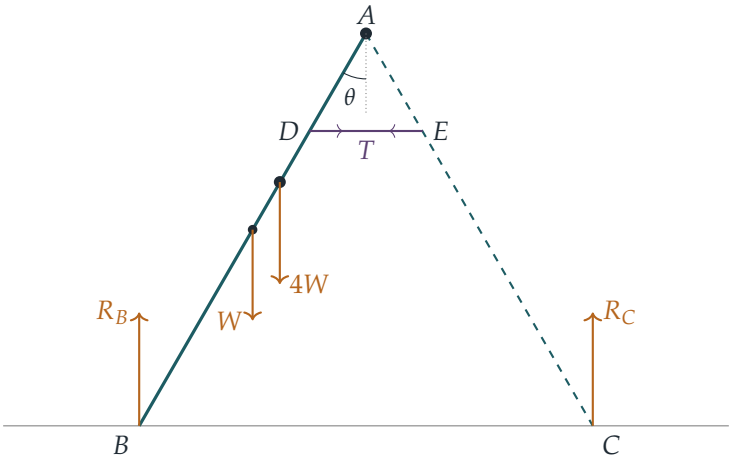
Because the floor is smooth, the only forces from the floor are vertical, so the whole ladder gives the reaction at C at once. The light leg AC then does the clever work: only three forces act on it (the floor reaction, the rope tension and the hinge force), so taking moments about the hinge A removes the unknown hinge force and links the rope tension to the reaction at C. Hooke's law ties the tension to the angle of the ladder, and that gives β as a function of the angle, which we maximise.

Solution

Setting up

We model AB as a uniform rod and AC as a light rod, both of length $4a$, with a smooth hinge at A, and the rope as light and elastic. The man is a particle. By symmetry ($AD = AE$ and $AB = AC$) the rope DE is horizontal. Let each leg make an angle θ with the vertical, with $0 < \theta < \frac{\pi}{2}$.

Since the floor is smooth, the reactions R_B and R_C at B and C are vertical. The weight W of AB acts at its midpoint, a horizontal distance $2a\sin\theta$ from B, and the man's weight $4W$ acts a horizontal distance $\beta a\sin\theta$ from B. The tension T pulls D and E towards each other.



The reaction at C

Taking moments about B for the whole step-ladder (the hinge force and the tension are internal, so they don't appear), with $BC = 8a\sin\theta$:

$$R_C \cdot 8a\sin\theta = W \cdot 2a\sin\theta + 4W \cdot \beta a\sin\theta \quad \Longrightarrow \quad R_C = \tfrac{1}{4}W(1+2\beta).$$

Resolving vertically, $R_B = 5W - R_C = \frac{1}{4}W(19-2\beta)$, which is positive since $\beta \leq 4$ (the rung is on the ladder). So the legs never lift off the floor.

The tension in the rope

The rope has length $DE = 2a\sin\theta$ and natural length $\frac{a}{4}$. If the rope were slack, $T = 0$ and moments about A for AC below would force $R_C = 0$, which is impossible. So the rope is stretched, $\sin\theta > \frac{1}{8}$, and Hooke's law gives

$$T = \frac{W\left(2a\sin\theta - \frac{a}{4}\right)}{a/4} = W(8\sin\theta - 1).$$

Equilibrium of the light leg AC

Ask yourself Why look at AC on its own rather than AB?

Leg AC is light, so only three forces act on it: R_C at C, the tension T at E and the hinge force at A. Moments about A remove the unknown hinge force. The tension is horizontal and acts at a vertical depth $a\cos\theta$ below A; the reaction R_C is vertical and acts a horizontal distance $4a\sin\theta$ from A. So

$$R_C \cdot 4a\sin\theta = T \cdot a\cos\theta.$$

Substituting for R_C and T:

$$\begin{aligned} W(1+2\beta)\sin\theta &= W(8\sin\theta - 1)\cos\theta, \\ \text{so} \quad 1+2\beta &= 8\cos\theta - \cot\theta, \\ \text{and} \quad \beta &= \tfrac{1}{2}\left(8\cos\theta - \cot\theta - 1\right). \end{aligned}$$

For each position β of the man, equilibrium is possible exactly when this equation has a solution θ (with $\sin\theta > \frac{1}{8}$). So the largest possible β is the maximum of $f(\theta) = 8\cos\theta - \cot\theta$.

Maximising

Differentiating,

$$f'(\theta) = -8\sin\theta + \operatorname{cosec}^2\theta = \frac{1 - 8\sin^3\theta}{\sin^2\theta}.$$

This is positive for $\sin\theta < \frac{1}{2}$ and negative for $\sin\theta > \frac{1}{2}$, since $\sin\theta$ increases on $(0, \frac{\pi}{2})$. So f has its greatest value when $\sin\theta = \frac{1}{2}$, that is, $\theta = \frac{\pi}{6}$, and then

$$f\left(\tfrac{\pi}{6}\right) = 8 \cdot \tfrac{\sqrt{3}}{2} - \sqrt{3} = 3\sqrt{3}, \qquad \beta_{\max} = \tfrac{1}{2}\left(3\sqrt{3} - 1\right) \approx 2.10.$$

This is less than 4, so the rung is on the ladder, and $\sin\theta = \frac{1}{2} > \frac{1}{8}$, so the rope is taut, as we assumed.

The height of the rung

The rung is a distance βa along BA from B, and BA makes angle θ with the vertical, so its height is

$$h = \beta a\cos\theta = \tfrac{1}{2}\left(3\sqrt{3} - 1\right) a \cdot \tfrac{\sqrt{3}}{2} = \frac{(9 - \sqrt{3})\,a}{4} \approx 1.82a.$$

Checks

- At $\theta = \frac{\pi}{6}$ with $W = a = 1$: $T = 8 \cdot \frac{1}{2} - 1 = 3$ and $R_C = \frac{1}{4}(1 + 3\sqrt{3} - 1) = \frac{3\sqrt{3}}{4} \approx 1.299$, and indeed $R_C \cdot 4\sin\theta = 2.598 = T\cos\theta$.
- A numerical check that solved the force and moment equations for both legs separately, and scanned θ over $(0, \frac{\pi}{2})$, found the largest β to be 2.09808 at $\theta \approx 30^\circ$, giving height $1.81699a$.
- Limiting cases make sense: as $\sin\theta \to \frac{1}{8}$ the rope goes slack and β would have to be negative, and as $\theta \to \frac{\pi}{2}$ the rope's moment arm $a\cos\theta$ vanishes, so again β must fall. A maximum in between is what we expect.

Common slips

- Taking the hinge force at A to be horizontal. It isn't, in general, because the loads on the two legs are different; moments about A avoid needing it at all.
- Using the whole length $DE = 2a\sin\theta$ as the extension. The extension is $2a\sin\theta - \frac{a}{4}$.
- Giving $\beta_{\max}$ (a distance along the ladder) instead of the height asked for, which needs the extra factor $a\cos\theta$.

Beyond STEP

This is a statics problem in which the configuration (θ) is not given but is set by the elastic rope, so equilibrium positions form a one-parameter family and the question becomes an optimisation over that family. At university this is the idea behind virtual work and potential energy methods: equilibria are stationary points of the potential energy, and asking which are stable leads to bifurcation and catastrophe theory. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good place to deepen the mechanics.

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